MIT ZARAGOZA
International Logistics Program

ZARAGOZA LOGISTICS CENTER

MZ 510

Analytical Methods

Sample Midterm

1. (25 points)

The Allen Corporation has three branch plants with excess production capacity. Fortunately, the corporation has a new product ready to begin production, and all three plants have this capability, so some of the excess capacity can be used in this way. This product can be made in three sizes--large, medium, and small--that yield a net unit profit of €420, €360, and €300, respectively. Plants 1, 2, and 3 have the excess capacity to produce 750, 900, and 450 units per day of this product, respectively, regardless of the size or combination of sizes involved.

The amount of available in-process storage space also imposes a limitation on the production rates of the new product. Plants 1, 2, and 3 have 13,000, 12,000, and 5,000 square feet, respectively, of in-process storage space available for a day's production of this product. Each unit of the large, medium, and small sizes produced per day requires 20, 15, and 12 square feet, respectively.

Sales forecasts indicate that if available, 900, 1,200, and 750 units of the large, medium, and small sizes, respectively, would be sold per day.

At each plant, some employees will need to be laid off unless most of the plant's excess production capacity can be used to produce the new product. To avoid layoffs if possible, management has decided that the plants should use the same percentage of their excess capacity to produce the new product.

Management wishes to know how much of each of the sizes should be produced by each of the plants to maximize profit.

Formulate a linear programming model for this problem.

2. (25 points)

Pure Food produces tomato sauce at five different plants. The capacity (in tons) of each plant is given in Table 2.1. The tomato sauce is stored at one of three warehouses. The per-ton cost of producing tomato sauce at each plant and shipping it to each warehouse is given on Table 2.2. Pure Food has four customers. The cost of shipping a ton of sauce from each warehouse to each customer is as given in Table 2.3. Each customer must be delivered the amount (in tons) of sauce given in Table 2.4 (these are annual demands).

In addition, Table 2.5 provides the fixed annual cost of operating each plant and warehouse. Formulate an integer program to minimize the total cost subject the supply and demand constraints and making sure that: (1) at most four plants can be used; and (2) customer 3 can only receive tomato sauce from one of the three warehouses.

Table 2.1 Capacities for Pure Food

	Plant				
	1	2	3	4	5
Tones	300	200	300	200	400

Table 2.2 Production and Shipping Costs for Pure Food

	To		
From	Warehouse 1	Warehouse 2	Warehouse 3
Plant l	€800	€1000	€1200
Plant 2	€700	€500	€700
Plant 3	€800	€600	€500
Plant 4	€500	€600	€700
Plant 5	€700	€600	€500

Table 2.3 Shipping Costs to Customers for Pure Food

	To			
From	Customer 1	Customer 2	Customer 3	Customer 4
Warehouse 1	€40	€80	€90	€50
Warehouse 2	€70	€40	€60	€80
Warehouse 3	€80	€30	€50	€60

Table 2.4 Demands for Pure Food

	Customer			
	1	2	3	4
Demand(tons)	200	300	150	250

Table 2.5 Fixed Costs for Pure Food

	Fixed Annual Cost
Plant l	€35,000
Plant 2	€45,000
Plant 3	€40,000
Plant 4	€42,000
Plant 5	€40,000
Warehouse 1	€30,000
Warehouse 2	€40,000
Warehouse 3	€30,000

3. (25 points)

BackSavers is a company that features a product line of backpacks of all shapes and sizes for students. They are considering offering some combination of three different models: the Jumbo, the Collegiate, and the Mini. They are trying to decide what quantities of each to produce.

Each full-time employee at BackSavers works a 40-hour week. By union contract, the number of full-time employees can never drop below 25. Part-time workers can also be hired. Each part-time employee works 20 hours per week. The union has imposed a restriction that there must be at least two full-time employees on the payroll for each part-time employee. Each full-time employee earns $16 per hour while each part-time employee earns $12 per hour.

All three types of backpacks are made out of the same rip-resistant nylon fabric. BackSavers has a long-term contract with a supplier and receives a 9000 square-foot shipment of the material each week. Each Jumbo requires 4 square feet of nylon; each Collegiate requires 3 square feet; each Mini requires 2 square feet.

There is a fixed cost associated with producing each type of backpack. If any Jumbos are produced, a weekly fixed cost of $7600 is incurred. If any Collegiates are produced, a weekly fixed cost of $9200 is incurred. If any Minis are produced, a weekly fixed cost of $6800 is incurred.

The sales forecasts indicate that at most 1600 Jumbos, 900 Collegiates, and 1200 Minis can be sold per week.

Each Jumbo requires 30 minutes of labor to produce, and generates a gross profit (not counting labor or fixed costs) of $28. Each Collegiate requires 45 minutes of labor to

produce and generates a gross profit of $32. Each Mini requires 35 minutes of labor and generates a gross profit of $24.

Management wishes to know what quantity of each of the three types of backpacks to produce per week, as well as how many full-time and part-time workers to employ. They would like to maximize their net profit – their gross profit from sales minus labor costs and fixed costs. Formulate a linear/integer programming program to solve the problem.

4. (25 points)

Company Scalers is a for-profit company and has a factory with production capacity of 500 units. It can do investments to expand capacity, however the higher the existing capacity is, the higher the cost of expanding it becomes. Expansion investments need to be done in multiples of $100, and the maximum budget for expansions is $1,000. The relationship between the amount of dollars invested and the resulting total capacity is given in Table 4.1.

Table 4.1:Investment in Production Capacity

Investment (in $'s)	Production Capacity
0	500
100	550
200	600
300	650
400	680
500	710
600	740
700	770
800	800
900	820
1,000	840

The company's product portfolio includes 3 products defined as: Alpha, Beta, and Gamma, with marginal profits (marginal revenue minus marginal production cost) of $10, $8, and $6, respectively. Each product requires 1 unit of capacity, and they can be sold in two markets: Spain and Portugal. Each market charges Scalers with a fixed fee of $50 per product introduced (independent of the quantity); Scalers is not willing to pay more than $250 in entering fees, and wants to send out at least 500 items in total. Demand for Alpha in Spain is at most 200 units, and the total quantity of Alpha produced must be equally distributed between both countries. Finally, at least 50 units of each product must be sent to Portugal.

Formulate an I.P. to learn how much money should Scalers invest in building additional capacity, and how many items of each product should be sent to Spain and to Portugal to maximize profit.

SOLUTION

Problem 1:

Let x_{ij} *= number of units produced at plant* i *of product* j *(*i *= 1, 2, 3;* j *= L, M, S).*

Maximize Profit = €$420(x_{1L} + x_{2L} + x_{3L})$ + €$360(x_{1M} + x_{2M} + x_{3M})$ + €$300(x_{1S} + x_{2S} + x_{3S})$

subject to

$x_{1L} + x_{1M} + x_{1S} \leq 750$
$x_{2L} + x_{2M} + x_{2S} \leq 900$
$x_{3L} + x_{3M} + x_{3S} \leq 450$
$20x_{1L} + 15x_{1M} + 12x_{1S} \leq 13{,}000$ *square feet*
$20x_{2L} + 15x_{2M} + 12x_{2S} \leq 12{,}000$ *square feet*
$20x_{3L} + 15x_{3M} + 12x_{3S} \leq 5{,}000$ *square feet*
$x_{1L} + x_{2L} + x_{3L} \leq 900$
$x_{1M} + x_{2M} + x_{3M} \leq 1{,}200$
$x_{1S} + x_{2S} + x_{3S} \leq 750$
$(x_{1L} + x_{1M} + x_{1S}) / 750 = (x_{2L} + x_{2M} + x_{2S}) / 900$
$(x_{1L} + x_{1M} + x_{1S}) / 750 = (x_{3L} + x_{3M} + x_{3S}) / 450$

and $x_{1L} \geq 0$, $x_{1M} \geq 0$, $x_{1S} \geq 0$, $x_{2L} \geq 0$, $x_{2M} \geq 0$, $x_{2S} \geq 0$, $x_{3L} \geq 0$, $x_{3M} \geq 0$, $x_{3S} \geq 0$.

Problem 2:

Let Pij = Number of tons produced at plant i and shipped to warehouse j,
Wjk = Number of tons sent from warehouse j to customer k,
Yi = 1 if plant i is used, Yi = 0 otherwise,
Zj = 1 if warehouse j is used, Zj = 0 otherwise.
Tj3 =1 if warehouse j is shipping to customer 3, 0 otherwise

MIN 35Y1+45Y2+40Y3+42Y4+40Y5
+30Z1+40Z2+30Z3++.8P11+P12+1.2P13+.7P21+.5P22+.7P23+.8P31+.6P32+.5P
33+.5P41+.6P42+.7P43+.7P51+.6P52+.5P53+.04W11+.08W12+.09W13+.05W14+.0
7W21+.04W22+.06W23+.08W2 4+.08W31+.03W32+.05W33+.06W34
SUBJECT TO

You can only produce if the plant is operating:
P11+P12+P13<=300Y1
P21+P22+P23<=200Y2
P31+P32+P33<=300Y3
P41+P42+P43<=200Y4
P51+P52+P53<=400Y5
Whatever goes into any warehouse needs to be equal to what is being shipped out of the same warehouse:
P11+P21+P31+P41+P51-W11-W12-W13-W14=0
P12+P22+P32+P42+P52-W21-W22-W23-W24=0
P13+P23+P33+P43+P53-W31-W32-W33-W34=0
If the warehouse is not open, then you cannot ship from that warehouse. The maximum amount shipped cannot be larger than the "total demand":
W11+W12+W13+W14<=900Z1
W21+W22+W23+W24<=900Z2
W31+W32+W33+W34<=900Z3
The demand at each customer must be met:
W11+W21+W31>=200
W12+W22+W32>=300
W13+W23+W33>=150
W14+W24+W34>=250
At most 4 plants can be used:
Y1+Y2+Y3+Y4+Y5<=4
Customer 3 can receive shipments from only one warehouse:
W13<=150 T13
W23<=150 T23
W33<=150 T33
T13+T23+T33=1
Pij, Wjk all non-negative
Yi,Zj,Tj3 all binary

Problem 3:

Following are the decision variables:

x_f : number of full - time employees to hire per week

x_p : number of part - time employees to hire per week

y_J : quantity of Jumbo backpacks to produce per week

y_C : quantity of Collegiate backpacks to produce per week

y_M : quantity of Mini backpacks to produce per week

$$z_J = \begin{cases} 1, \text{if any Jumbos are produced} \\ 0, \text{otherwise} \end{cases}$$

$$z_C = \begin{cases} 1, \text{if any Collegiates are produced} \\ 0, \text{otherwise} \end{cases}$$

$$z_M = \begin{cases} 1, \text{if any Minis are produced} \\ 0, \text{otherwise} \end{cases}$$

Therefore, we have the following integer program:

$$Max\ (28y_J + 32y_C + 24y_M) - (640x_f + 240x_p) - (7600z_J + 9200z_C + 6800z_M)$$

subject to :

$$4y_J + 3y_C + 2y_M \le 9000$$

$$x_f \ge 25$$

$$x_f \ge 2x_p$$

$$30y_J + 45y_C + 35y_M \le 2400x_f + 1200x_p$$

$$y_J \le 1600z_J$$

$$y_C \le 900z_C$$

$$y_M \le 1200z_M$$

$$x_f, x_p, y_J, y_C, y_M \ge 0$$

z_J, z_C, z_M binary

Problem 4:

Decision Variables:

d_i: quantity of "\$100 investments done" that yield a marginal increased capacity of i

$i \in I = \{50, 30, 20\}$

$x_{p,c}$: quantity of products p shipped to country c

$y_{p,c}$: 1 if at least one product p is shipped to country c; 0 else.

$p \in P = \{\alpha, \beta, \gamma\}; c \in C = \{S, T\}$, where S=Spain; P=Portugal.

Objective Function:

$$Z = \max \sum_c (10x_{\alpha,c} + 8x_{\beta,c} + 6x_{\gamma,c}) - 50 \sum_p \sum_c y_{p,c} - 100 \sum_i d_i$$

Constraints:

$$0 \leq d_{50} \leq 3$$

$$0 \leq d_{30} \leq 5$$

$$0 \leq d_{20} \leq 2$$

$$\sum_p \sum_c x_{p,c} \geq 500$$

$$\sum_p \sum_c x_{p,c} \leq 500 + 50d_1 + 30d_2 + 20d_3$$

$$\sum_p \sum_c y_{p,c} \leq 5$$

$$x_{p,c} \leq M y_{p,c}, \qquad \forall\, p \in P, c \in C$$

$$x_{\alpha,S} \leq 200$$

$$x_{\alpha,S} = x_{\alpha,T}$$

$$x_{p,T} \geq 50, \qquad \forall\, p \in P$$

$$x_{p,S} \geq 0, \qquad \forall\, p \in P$$

$$y_{p,c} \text{ binary} \qquad \forall\, p \in P, c \in C$$

$$d_i \text{ integer}, \qquad \forall\, i$$